FINAL PRESENTATION

FAIR MACHINE LEARNING FOR ANALYZING INCOME OPPORTUNITIES USING THE 1994 CENSUS DATA

Introduction to DataScience

LECTURER: MR. QUÁCH ĐÌNH HOÀNG

DATASET:
Income Adult

GROUP 5:
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23110060 - TRẦN HUỲNH XUÂN THANH

Contentlist

1. Introduction & Goal of Study
2. Data Description
3. EDA
4. Feature Engineering
5. Model Building
6. Results
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Goal of the study

- **Feature Impact:** Which factors (education, age, occupation) have the strongest influence on income?
- **Model Performance:** Which preprocessing techniques and models achieve the best performance and stability on imbalanced data?
- **Bias & Explainability:** Does the model exhibit bias with respect to gender or race? How can predictions be explained using SHAP or LIME?
- **Controlled Analysis:** When holding working hours and education constant, how do gender and race affect income?

Motivation: Personal income reflects key economic factors and helps understand social inequality. At the same time, this problem provides a practical setting to apply machine learning techniques and build interpretable and fair predictive models.

Analysis Approach & Evaluation

Problem Handling: Check for class imbalance in the dataset

- Clean and preprocess the Adult Census dataset
- Handle missing values and remove redundant features
- Apply feature engineering and feature selection

Train and compare ML models: Logistic Regression, Random Forest, LightGBM, XGBoost

Evaluation Metrics: Use F1-score (balancing precision and recall), AUC-ROC, MCC, and Stratified K-Fold cross-validation to evaluate and compare models

Explainability & Fairness (XAI): Apply techniques such as SHAP and LIME, along with fairness metrics

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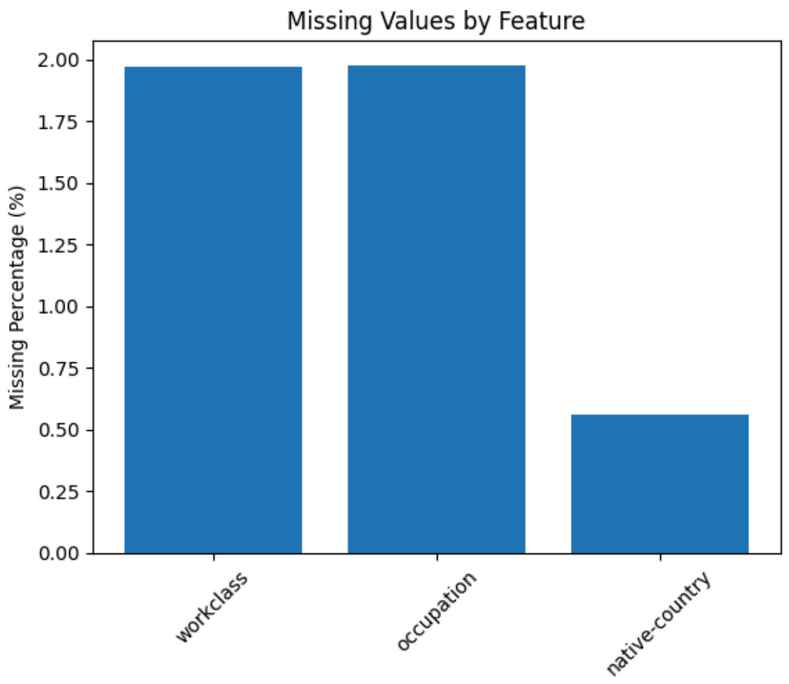
Dataset description

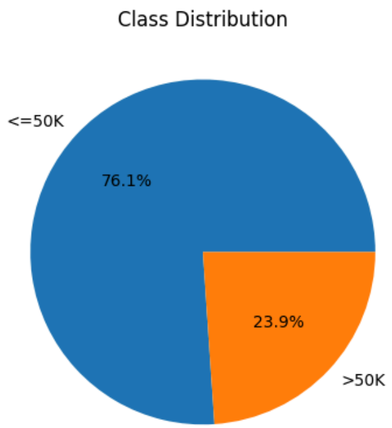
Source: Extracted from the 1994 U.S. Census Bureau database by Barry Becker and Ronny Kohavi

Scale: 48,842 independent observations (76% training, 24% testing).

Number of Variables: 15 variables (14 features and 1 target variable).

Data Issues: Approximately 3,620 rows contain missing values (about 7% of the dataset)





Features Description

Numerical Features:

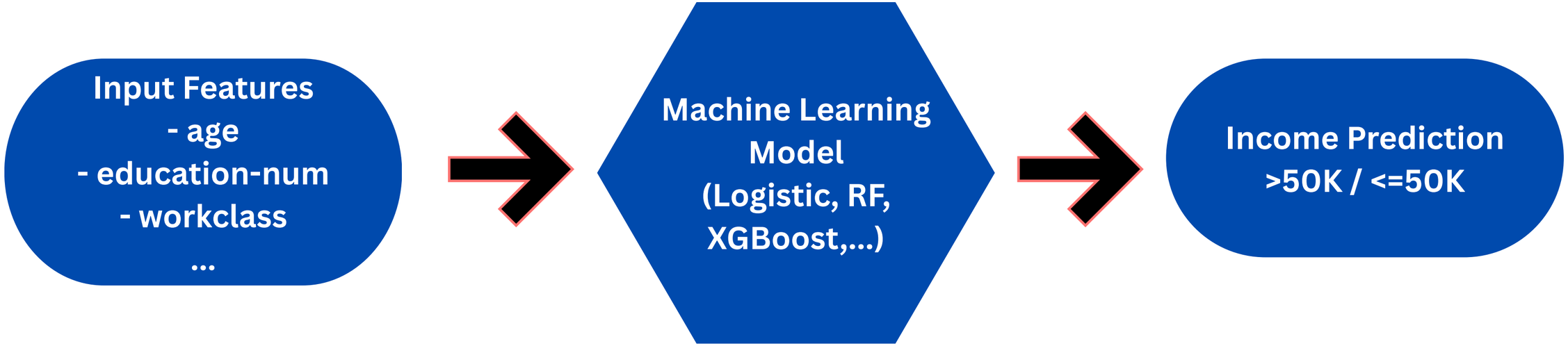
- age, education-num, hours-per-week (personal characteristics and work intensity)
- capital-gain, capital-loss (Investment-related activities)
- fnlwgt (Census sampling weight)

Categorical Features:

- workclass, education, marital-status, occupation, relationship, race, native-country (Reflect social and socio-economic characteristics), sex (gender)

Input và Output of Problem

Input/Feature X: 14 categorical và numberic variable



Output/Target Y: Binary variable "income"

- Y=1: Income >50.000 USD
- Y=0: Income <=50.000 USD

Moreover, outputs also answer these goal questions

- Model metrics.
- Fairness Metrics (Disparate Impact), SHAP Waterfall Plot
- SHAP Summary Plot, Feature Importance (Bar chart)...

Limitations of dataset

- **Outdated data (1994):** Income patterns may no longer reflect current conditions
- **Binary classification:** Oversimplifies the problem and loses information about income levels
- **Missing important features:** Lacks variables such as soft skills, location, assets, and social connections
- **Imbalance and missing data:** Requires handling (e.g., SMOTE, class weighting, imputation)
- **Risk of historical bias:** The model may learn and amplify existing social biases

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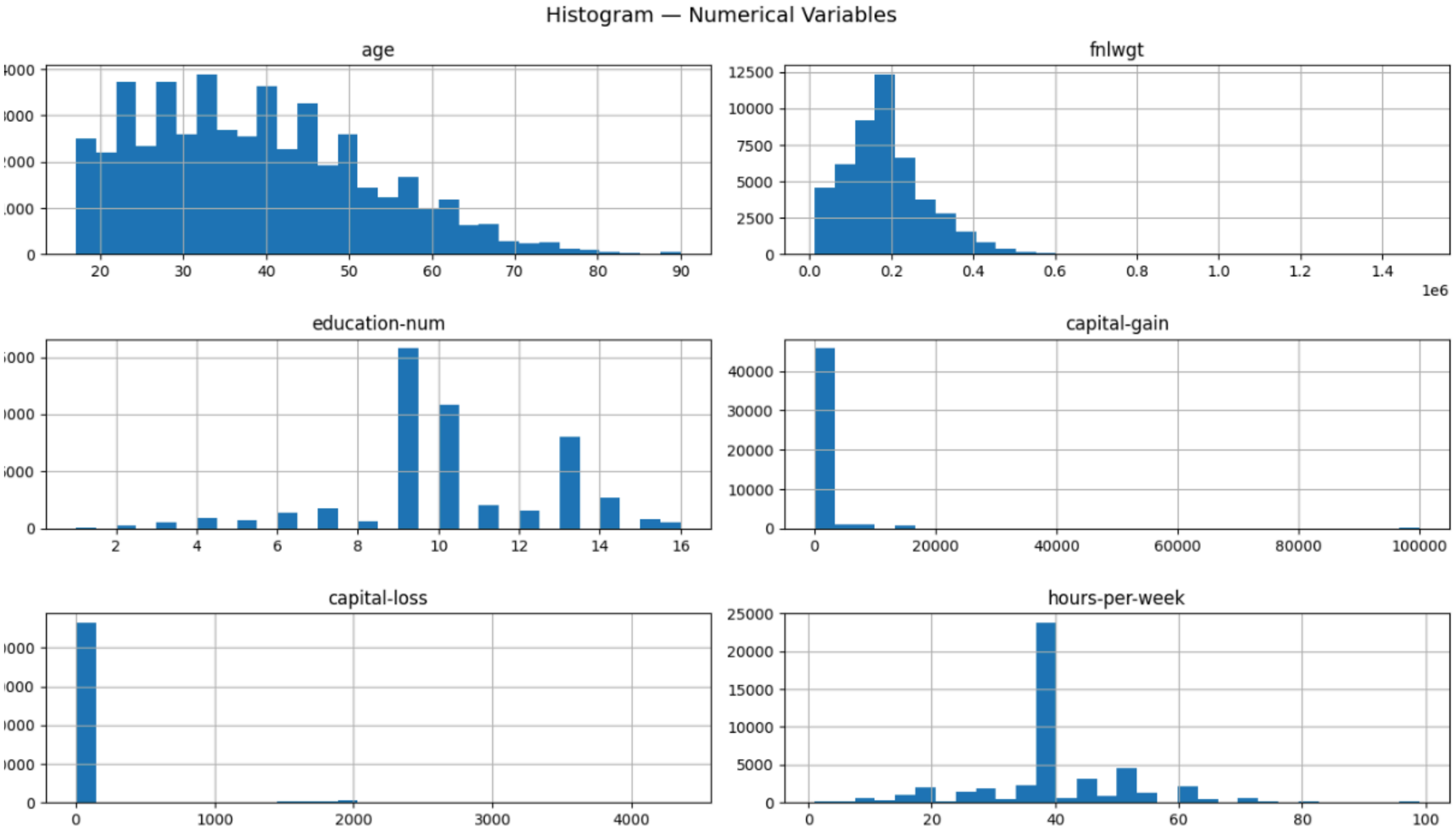
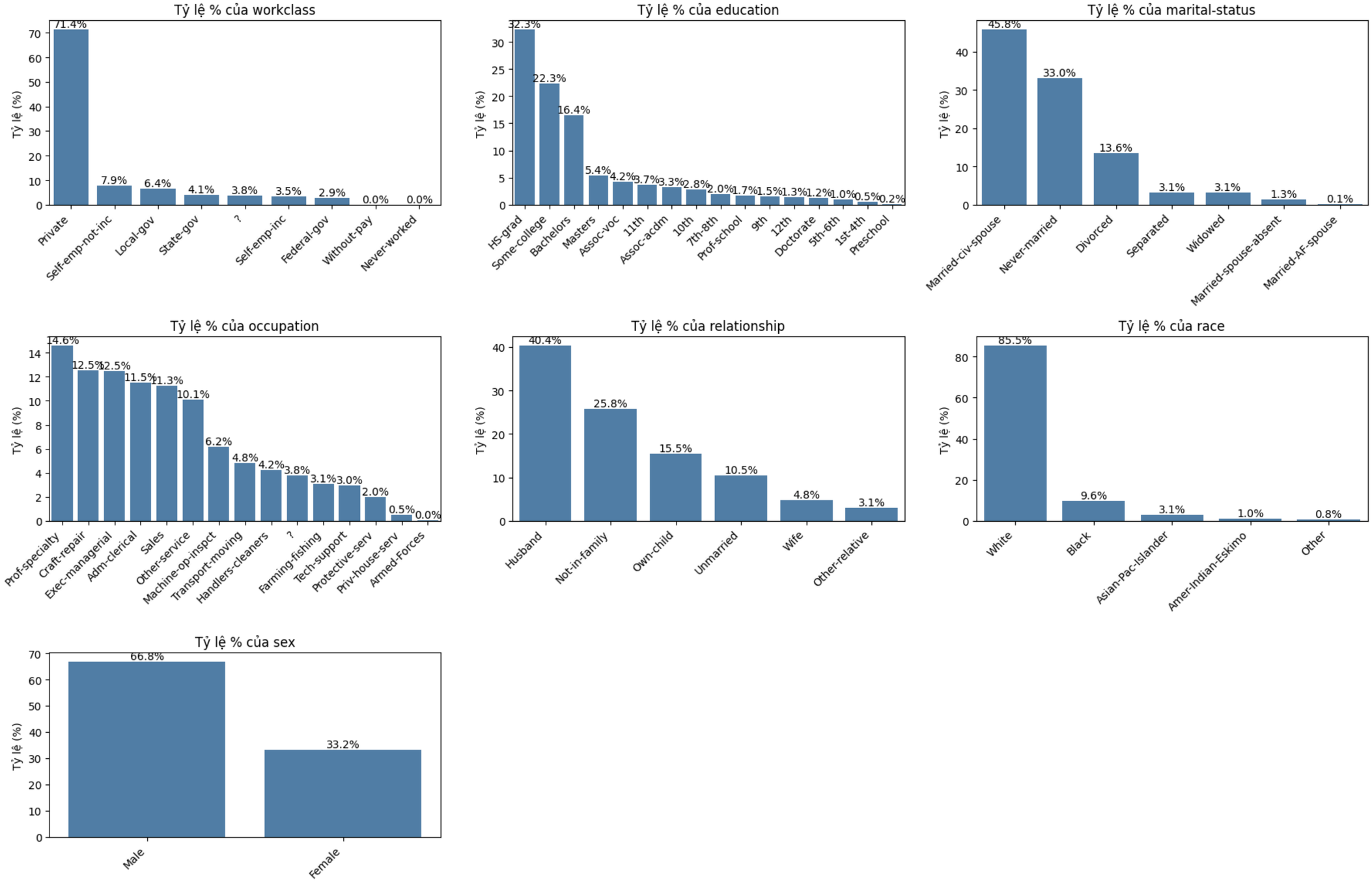
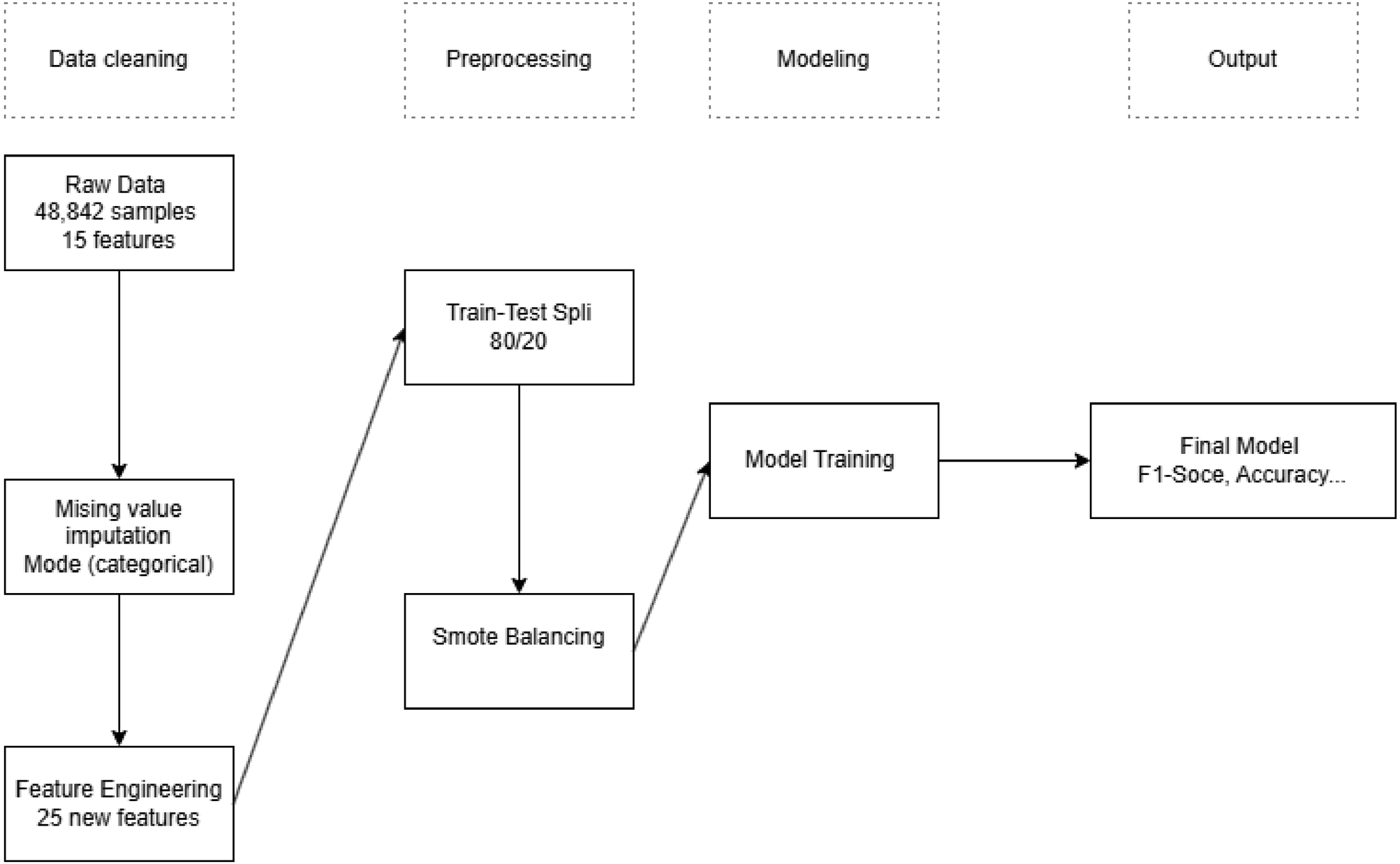
EDA


Histogram — Numerical Variables
age
fnlwgt
1e6
education-num
capital-gain
capital-loss
hours-per-week

Category Proportions



Processing Pipeline



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Feature Engineering Summary

From Original Variables

New Feature	Method
country_income_group	Mapping + Ordinal enc.
occupation_group	Grouping + CatBoost enc.
workclass (encoded)	LOO encoding
married_flag	Binary transformation
sex_binary	Binary encoding

One-Hot Encoded Features

Feature Group	From
relationship_* (6 vars)	relationship
race_* (5 vars)	race

Econometric Interaction Features

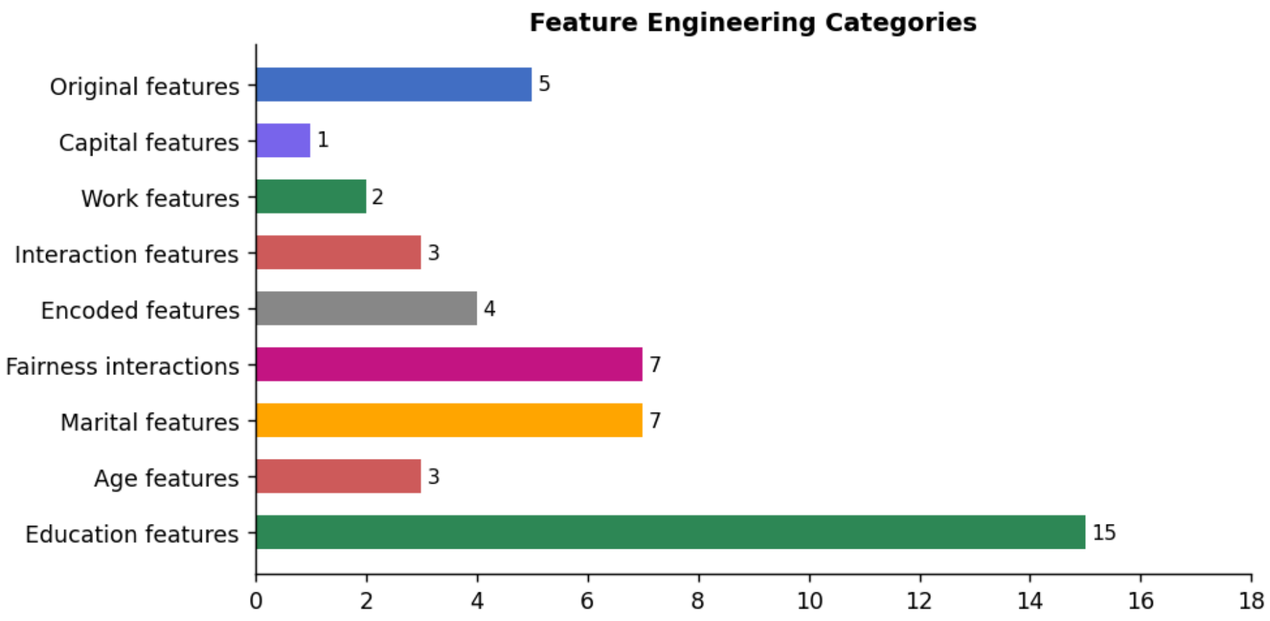
Feature	Formula
human_capital	age × education-num
household_labour	hours-per-week × married_flag
net_capital	capital-gain – capital-loss

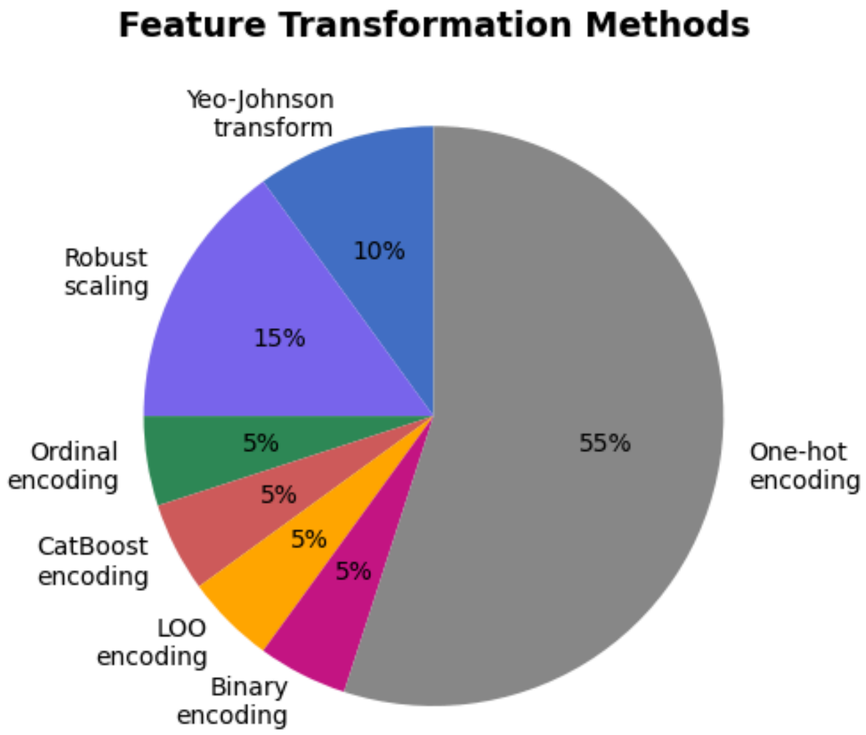
Fairness Interaction (Race)

Feature Group	Idea
edu_x_race_* (5 vars)	Edu return by race
hours_x_race_* (5 vars)	Working hours by race
capital_x_race_* (5 vars)	Capital return by race

Create interaction features between race and key socio-economic variables (education, hours, capital) to analyze fairness.
Form three feature groups: edu_x_race_*, hours_x_race_*, and capital_x_race_*.

Feature Engineering





►**Feature engineering primarily increases the number of features through interaction terms and fairness-related transformations, rather than simple encoding alone.**

►**A variety of encoding and scaling techniques are applied, but most new features are generated from one-hot encoding and interaction features.**

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Machine Learning Models Used

To address this binary classification problem, the following models are applied:

- Logistic Regression
- Random Forest Classification
- LightGBM
- XGBoost Classification

- The models are selected to balance interpretability and predictive performance. Logistic Regression serves as a baseline, while Random Forest, XGBoost, and LightGBM capture non-linear relationships.
- This combination ensures accurate prediction and reliable feature importance analysis.

Logistic Regression & Random Forest config

Parameter	Value
Penalty	L2
C	1
Solver	liblinear
Max Iter	2000

Logistic Regression

Parameter	Value
N Estimators	300
Max Depth	8
Learning Rate	0.05
Subsample	0.8
Regularization	L1 = 0.1, L2 = 1.0

Random Forest

► Model parameters are selected based on configurations reported in prior research to ensure academic rigor and comparability

►These configurations have been validated on similar tasks and are therefore expected to provide stable performance with the proposed data exploration and preprocessing approaches

XGBoost & LightGBM config

Parameter	Value
N Estimators	300
Max Depth	12
Min Samples Split	10
Min Samples Leaf	5
Max Features	sqrt

XGBoost

Parameter	Value
N Estimators	300
Max Depth	8
Learning Rate	0.05
Subsample	0.8
Colsample Bytree	0.8
Regularization	L1 = 0.1, L2 = 1.0

LightGBM

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Metrics

Model	Accuracy	Balanced Accuracy	Precision	Recall	F1-Score	AUC	MCC	Overfitting Gap
LR	0.837752	0.743116	0.701014	0.561591	0.623605	0.890605	0.527007	0.001753
RF	0.862012	0.760528	0.798913	0.565868	0.662494	0.916498	0.592529	0.023205
XGB	0.866721	0.787075	0.768394	0.634303	0.694939	0.922811	0.615264	0.032789
LGBM	0.874603	0.797093	0.789995	0.648417	0.712239	0.929924	0.63811	0.033435

Result after not handling imbalance

Model	Accuracy	Balanced Accuracy	Precision	Recall	F1-Score	AUC	MCC	Overfitting Gap
LR	0.794861	0.806531	0.547149	0.828914	0.659184	0.890088	0.544109	0.000267
RF	0.818917	0.82586	0.584799	0.839179	0.689268	0.916004	0.585594	0.022032
XGB	0.847989	0.821665	0.654922	0.771172	0.708309	0.920376	0.610142	0.025858
LGBM	0.859863	0.830789	0.682486	0.775021	0.725816	0.927754	0.634509	0.039002

Result after applying smote

Metrics

Model	Accuracy	Balanced Accuracy	Precision	Recall	F1-Score	AUC	MCC	F1 Gap	Optimal Threshold
XGBoost	0.8641	0.8286	0.6984	0.7605	0.7281	0.9302	0.6387	0.0247	0.753
LightGBM	0.8701	0.8223	0.7277	0.7305	0.7291	0.9306	0.6437	0.018	0.537
RandomForest	0.8464	0.8222	0.6499	0.7759	0.7074	0.9201	0.6086	0.0277	0.564

DOI: 10.17148/IJARCCE.2025.14801 using optimal threshold

Model	Accuracy	Balanced Accuracy	Precision	Recall	F1-Score	AUC	MCC	F1 Gap
LR	0.794861	0.806531	0.547149	0.828914	0.659184	0.890088	0.544109	0.000267
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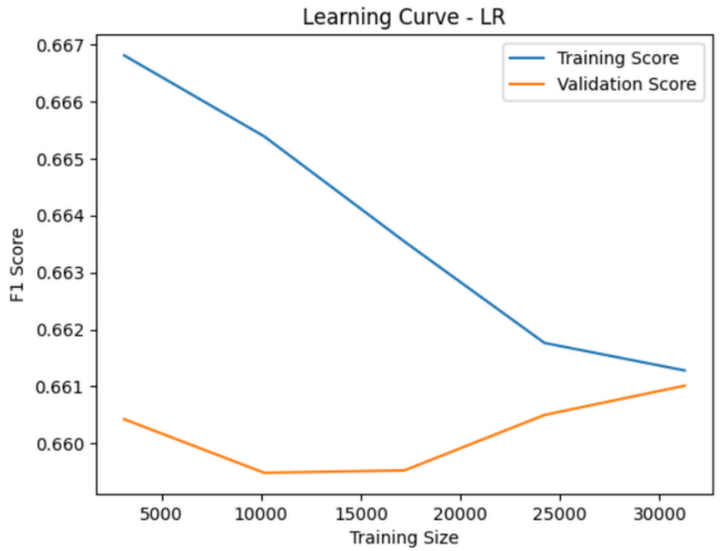
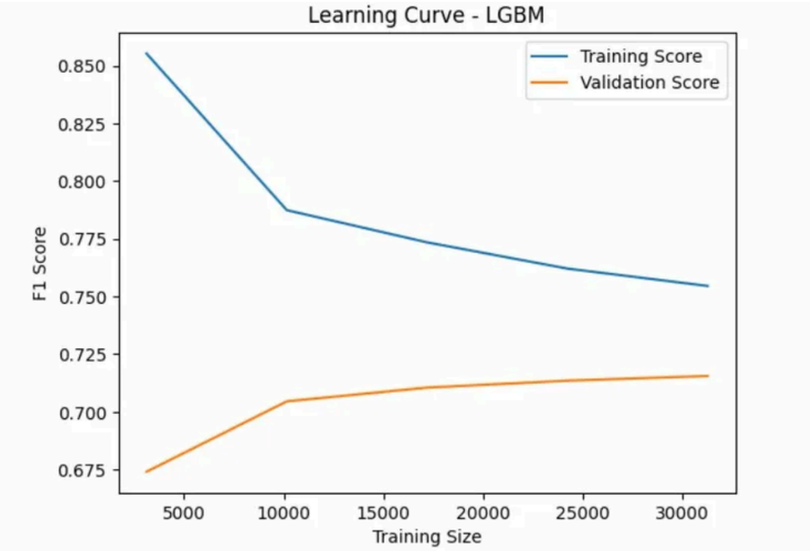
LightGBM	0.859863	0.830789	0.682486	0.775021	0.725816	0.927754	0.634509	0.039002

Result after applying smote

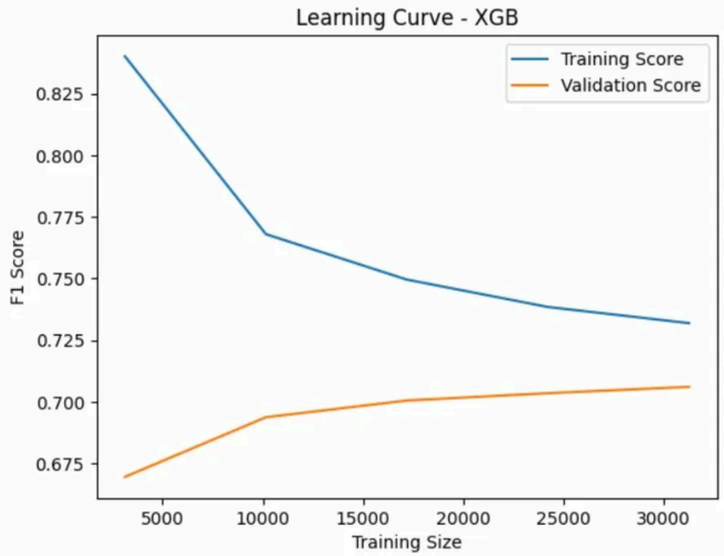
Analysis Learning Curves — SMOTE



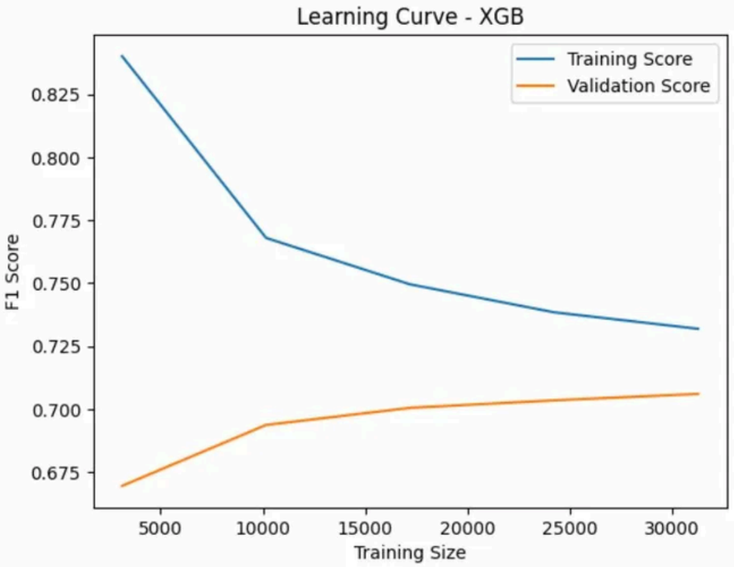
LR Final Overfitting Gap: 0.0003



LGBM Final Overfitting Gap: 0.0390

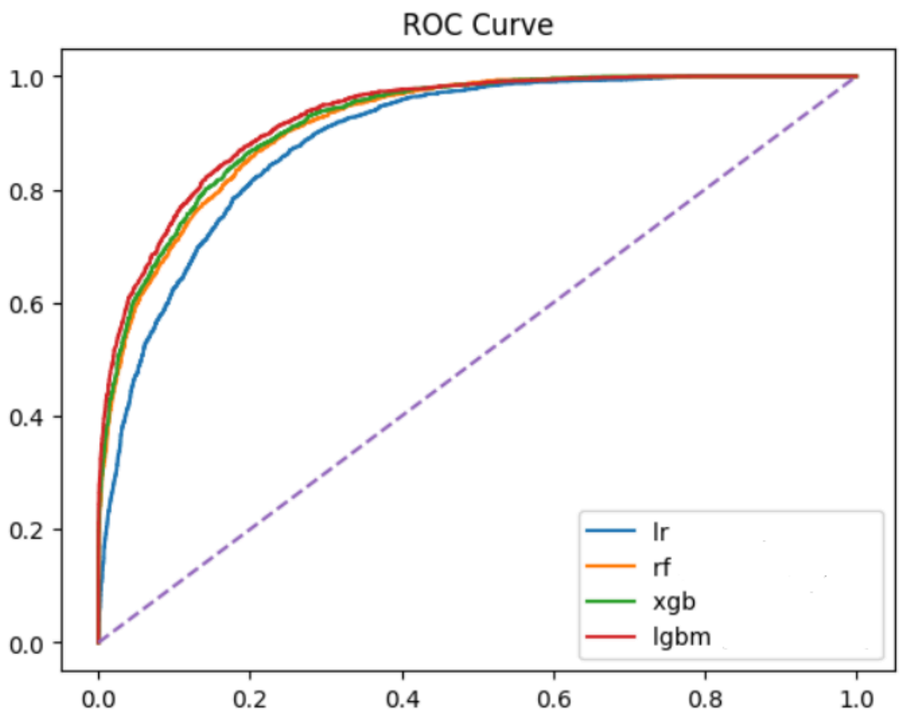


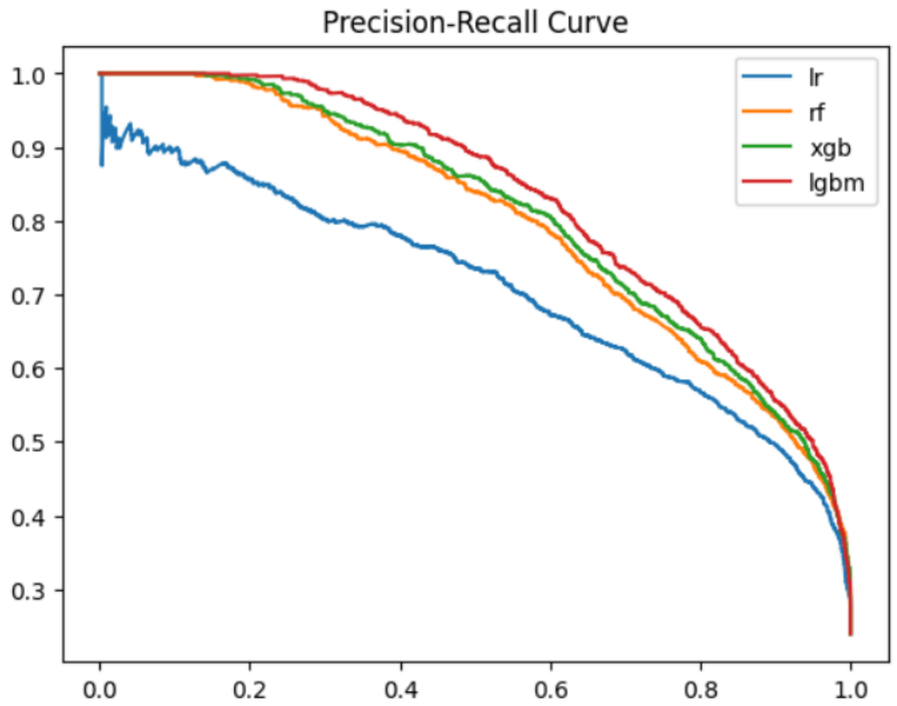
XGB Final Overfitting Gap: 0.0259



XGB Final Overfitting Gap: 0.0259

ROC & Precision-Recall Curve

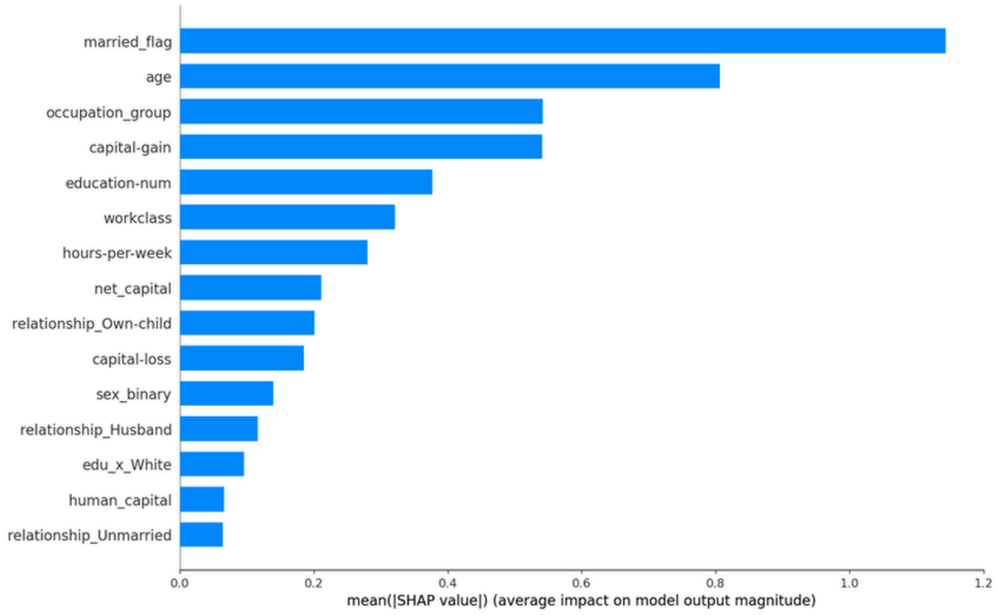


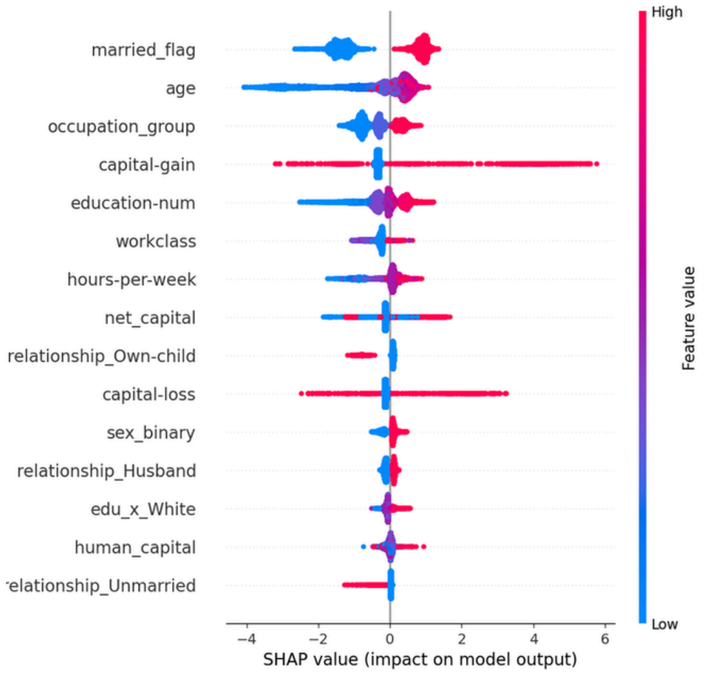


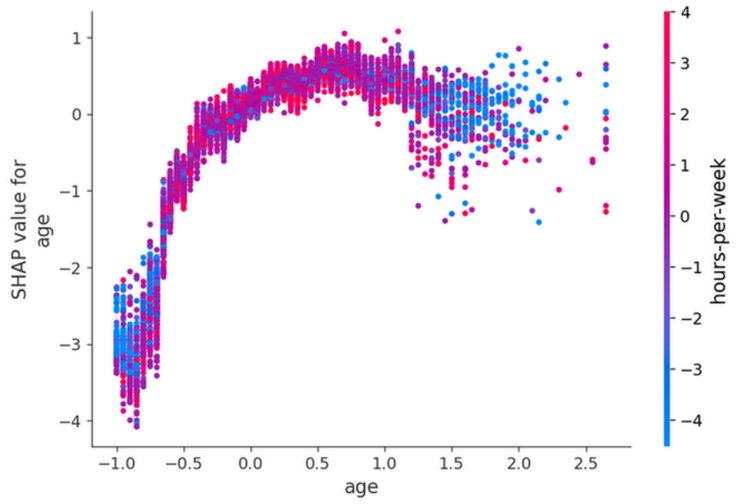
►LightGBM creates very sharp decision boundaries, which helps minimize the overlap between the two income groups, as reflected by the low number of false positives

►With the highest PR curve, LightGBM demonstrates that when it predicts an individual to have an income greater than 50k, the prediction is the most reliable, with the highest likelihood of being correct.

SHAP Analysis

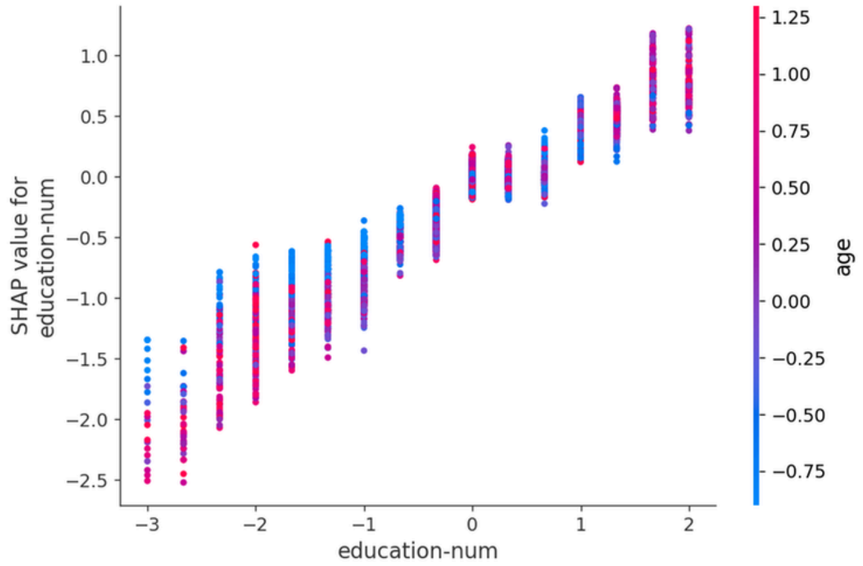


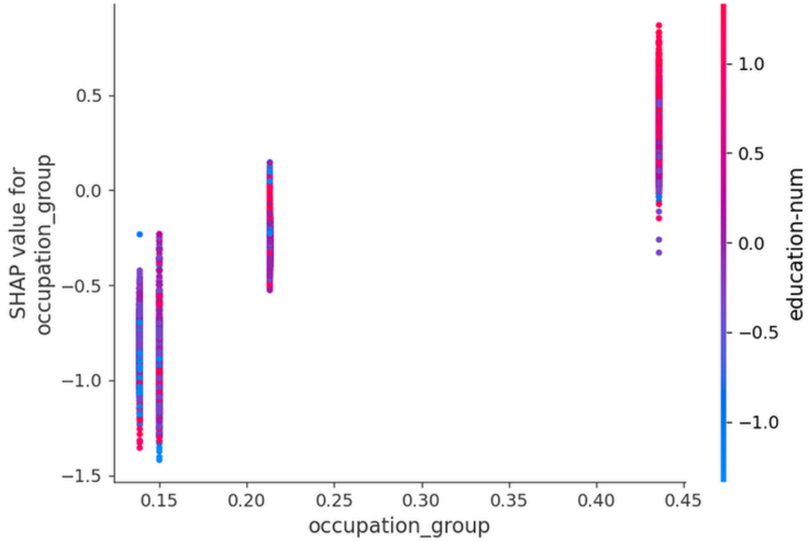


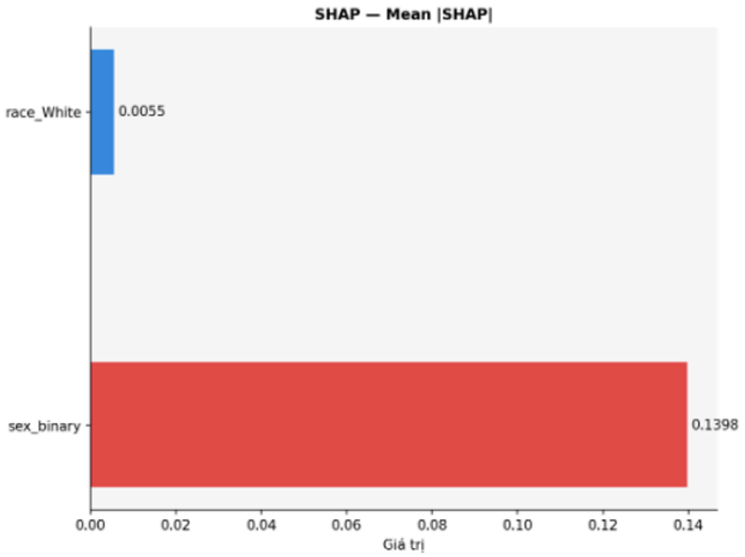


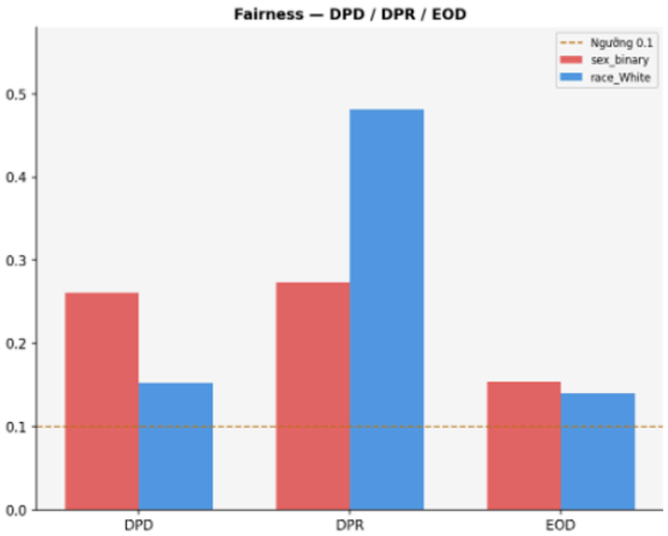
- **FEATURE IMPACT RANKING:** MARRIED_FLAG HAS THE STRONGEST INFLUENCE (~1.14), FOLLOWED BY AGE (~0.80), WHILE EDUCATION-NUM IS MUCH WEAKER (~0.38). AGE OUTWEIGHS BOTH OCCUPATION AND EDUCATION.
- **NON-LINEAR EFFECTS & INTERACTIONS:** AGE SHOWS AN INVERTED-U PATTERN (YOUNG: SHAP ≈ -4, PEAKS AT MIDDLE AGE, THEN DECLINES); MARRIED_FLAG CONSISTENTLY BOOSTS PREDICTIONS; STRONG INTERACTION WITH HOURS-PER-WEEK.
- **CORE INSIGHT:** THE MODEL DOES NOT LEARN TRUE “ABILITY” BUT LARGELY REFLECTS REAL-WORLD INCOME DISTRIBUTION, WITH AGE AND MARITAL STATUS AS THE DOMINANT FACTORS.

Fairness Explaination



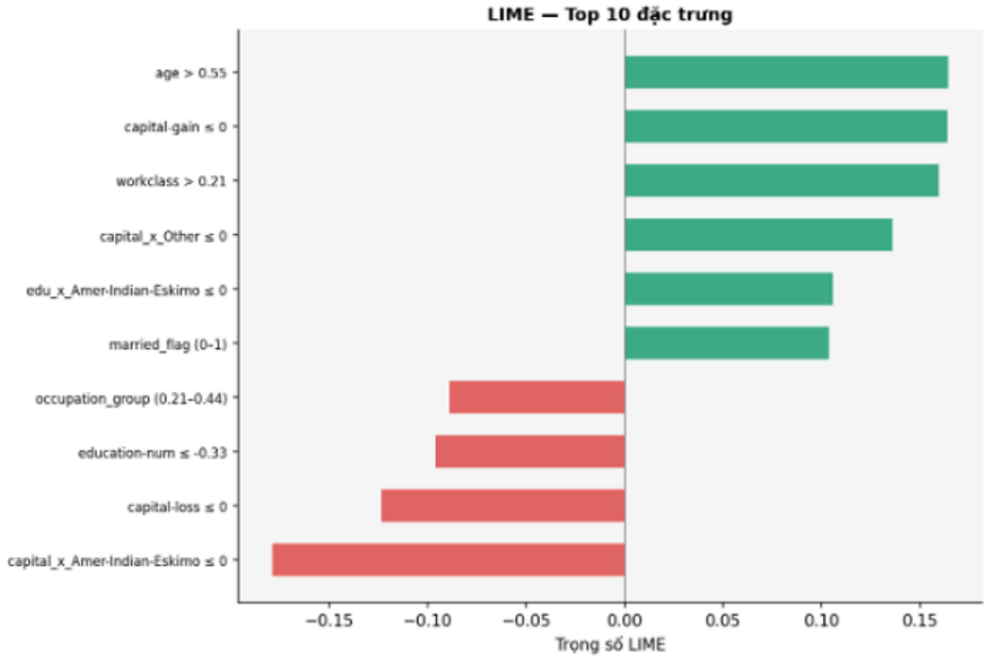


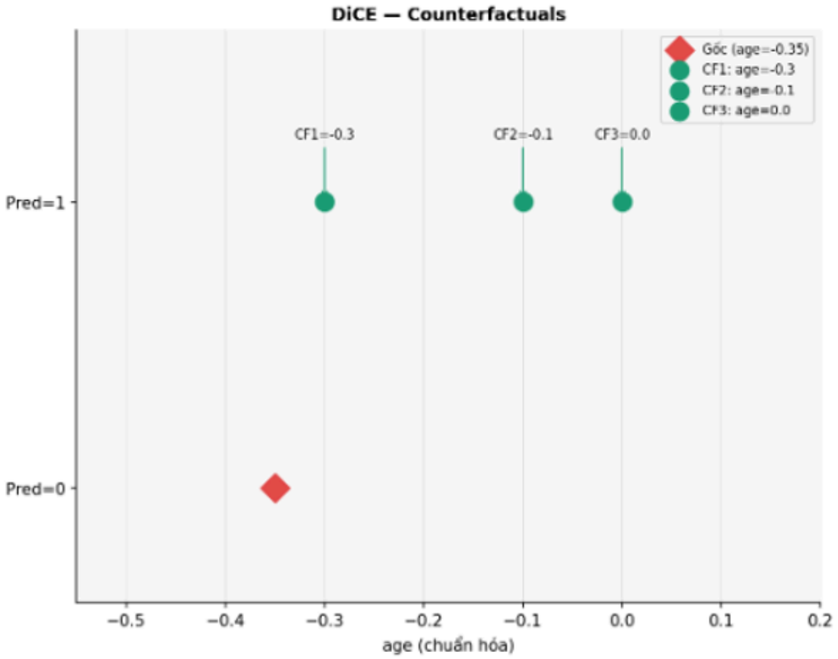


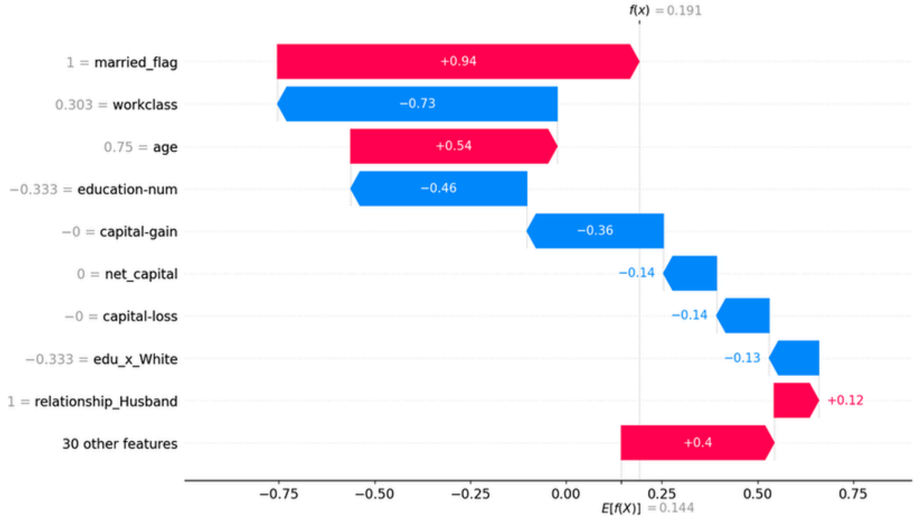


- Fairness analysis shows strong bias across gender and race (high DPD, DPR ~0.27, EOD above threshold).
- However, SHAP indicates sensitive features have low direct impact → the model learns bias through proxy variables (e.g., married_flag, occupation_group, relationship).
- The model not only reflects historical inequality but reinforces it—bias is embedded in the data structure, not just sensitive features.

LIME Analysis

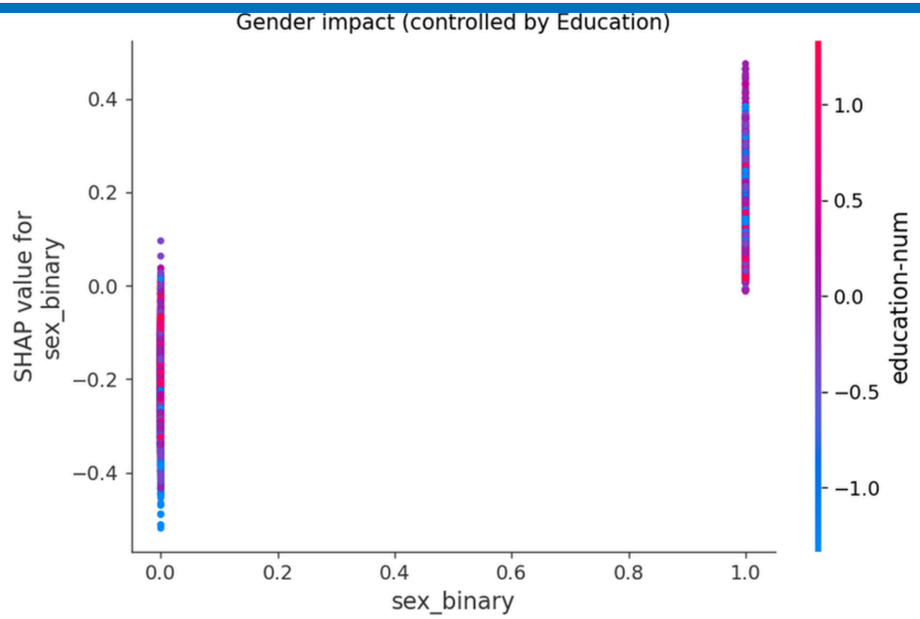


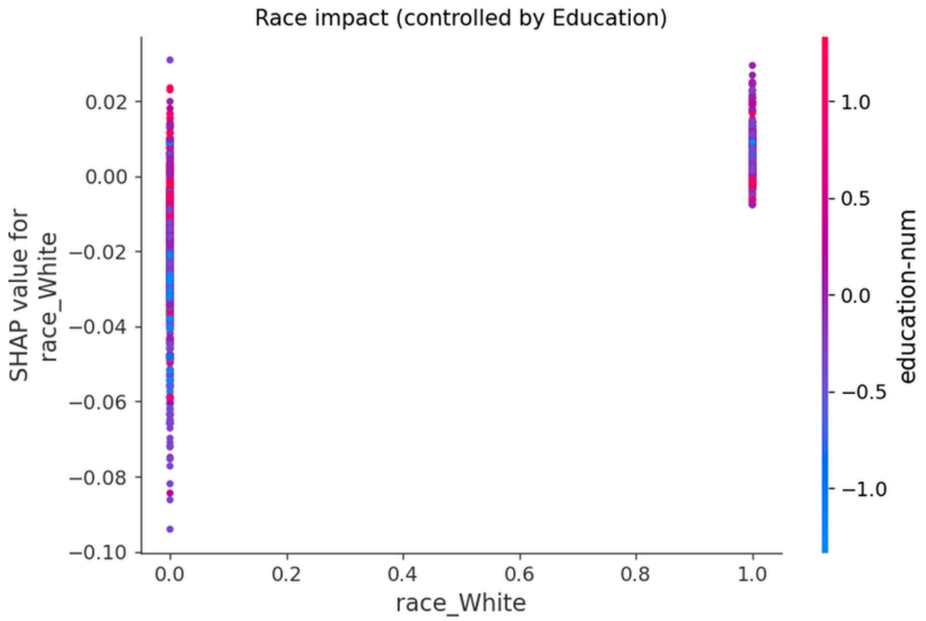


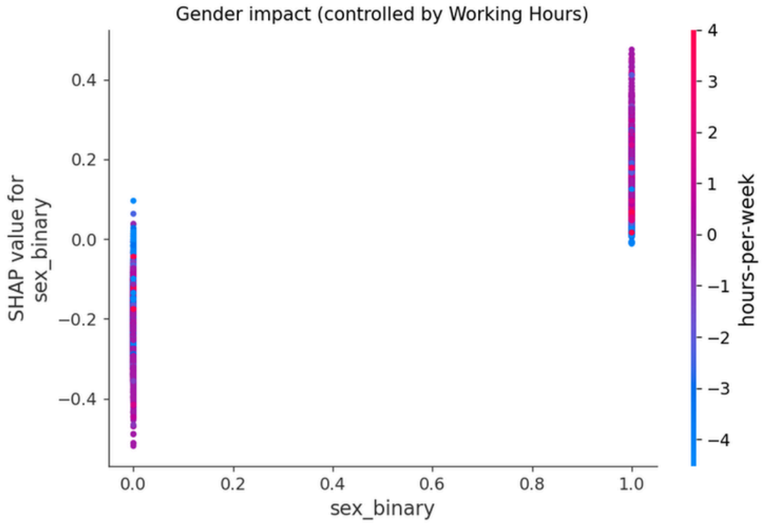


- LIME shows a trade-off between positive and negative factors: married_flag (+0.94) and age (+0.54) push predictions up, while workclass (-0.73), education-num (-0.46), and capital-gain (-0.36) pull them down.
- This indicates the model prioritizes age and social structure over education.
- DiCE reveals that a small increase in age (from -0.35 to ~ -0.30) can flip the outcome → a sharp decision threshold.

Fairness Explaination







- EVEN AFTER CONTROLLING FOR EDUCATION AND WORKING HOURS, BIAS STILL PERSISTS:

+ Gender: same education/hours → one group gets negative SHAP, the other positive

+ RACE: SAME EDUCATION → WHITE GROUP STILL HAS HIGHER SHAP

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Conclusion

- The analysis shows that the model does not purely learn individual capability, but mainly captures the social structure of income distribution. Factors like age, marital status, and occupation have stronger influence than education.
- Among key variables, age is the most important driver, with clear nonlinear effects, while education has significantly weaker impact.
- In terms of modeling, feature engineering + SMOTE + LightGBM provides the best performance and stability, while remaining interpretable via SHAP and LIME.
- Fairness analysis reveals clear bias in gender and race, even though sensitive features have low direct importance. This is due to proxy variables (e.g., occupation, marital status) indirectly encoding these attributes.
- Even after controlling for education and working hours, disparities persist, indicating that inequality is embedded in the data structure, not individual ability.
-